Modern Cryptography
Homework 1

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Question 1 (Encryption)**.** Prove the following:

(a) (15 points) There is a perfectly secret encryption scheme that reveals 99% of the key.

(b) (15 points) There is an encryption scheme that is not perfectly secure, yet no adversary can guess the key with probability better than $2^{-\lambda}$, where λ is the size of the key.

Question 2 (Negligible Functions)**.** Recall that a non-negative function $f : \mathbb{N} \to \mathbb{R}$ is negligible if it decreases faster than any inverse polynomial. For each of the following functions, state whether they are negligible or not and give a brief explanation. Here log denotes the base-2 logarithm, negl denotes some negligible function and poly denotes some polynomial.

1. (5 points) n^{-100}
2. (5 points) $2^{-100 \log n}$
3. (5 points) $n^{-\log n}$
4. (5 points) $\text{poly}(n) \cdot \text{negl}(n)$

Question 3 (Indistinguishability)**.** (20 points) Let D_0, D_1 be two distributions and let A be an algorithm that outputs a single bit. We say that A distinguishes the distributions D_0, D_1 with probability ε if

$$\left| \Pr_{x \leftarrow D_0}[A(x) = 1] - \Pr_{x \leftarrow D_1}[A(x) = 1] \right| = \varepsilon.$$

Now suppose that a challenger C plays the following game with A. First, C picks a random bit $b \leftarrow \{0, 1\}$ and then it samples an $x \leftarrow D_b$ and gives this x to A. Finally, A gives back to C a bit $A(x)$. Show that A returns the correct bit b with probability $1/2 + \varepsilon/2$ if and only if A can distinguish the distributions D_0, D_1 with probability ε. Equivalently,

$$\left| \Pr_{x \leftarrow D_0}[A(x) = 1] - \Pr_{x \leftarrow D_1}[A(x) = 1] \right| = \varepsilon \iff \left| \Pr_{\substack{b \leftarrow \{0,1\} \\ x \leftarrow D_b}}[A(x) = b] - \frac{1}{2} \right| = \frac{\varepsilon}{2}$$

Question 4. Let p be an odd prime.

1. (15 points) Find an efficient algorithm to check if a given $a \in \mathbb{Z}_p^*$ is a quadratic residue.
2. (15 points) Describe an efficient algorithm to check if a given $g \in \mathbb{Z}_p^*$ is a generator. Assume that the algorithm is also given a factorization of $p - 1$. (It is not known how to perform this task efficiently without this factorization.)